

# SUBSTITUTION MODELS

## WEEK 37 RECAP + PAPER DISCUSSION



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EVOLUTIONARY THINKING  
29. AUGUST 2025

EMMA FEMKE DIEPEVEEN  
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# WHAT ARE WE DOING TODAY

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primate of the week

Recap of nucleotide substitution models

Paper discussion on on different substitution models

Menti with things you need more explanation about



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# PRIMATE OF THE WEEK

## Mandrill

*Mandrillus sphinx*

Occurs in Africa

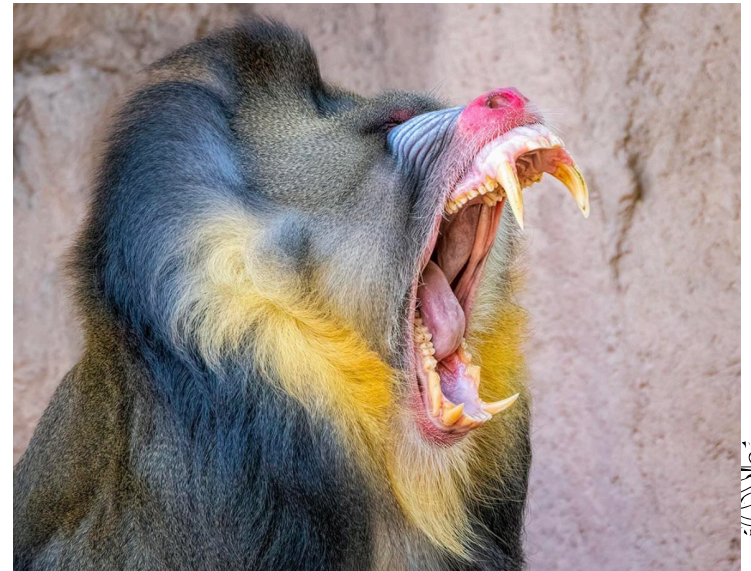
Largest monkey species in the world

**Sexual dimorphism:** Males are larger than females and more colourful

Colour indicates their testosterone level and also their status.

Females tend to prefer more colourful males.

When mandrills become excited, the colouring on their buttocks further brightens.



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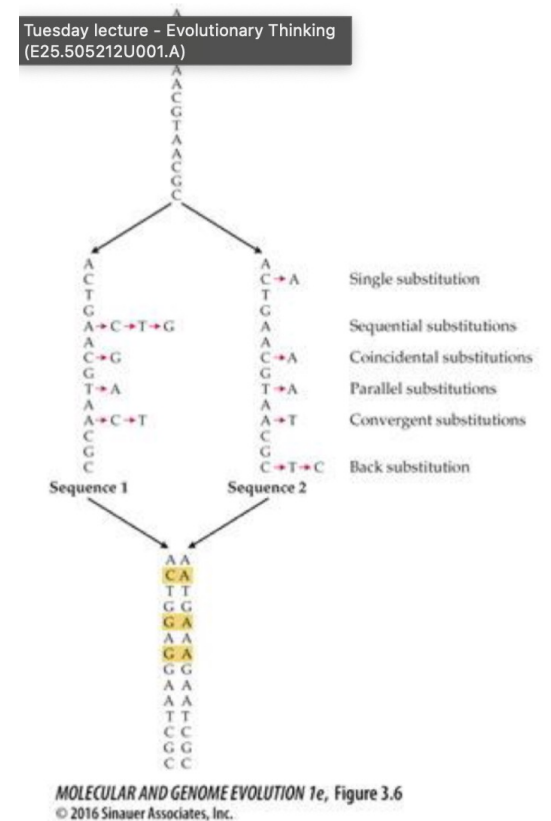


# NUCLEOTIDE SUBSTITUTIONS

Divergent sequences

AGATGCTAGCATCGACTAGCARCAGCTGACCCGCGCGCGCAT  
AGAT**AA**TAGCATCGACTAGCARCAG**C**GGACCCGCGCGCGCAC**C**

You could just count how many difference there are but that's wrong



# NUCLEOTIDE SUBSTITUTIONS

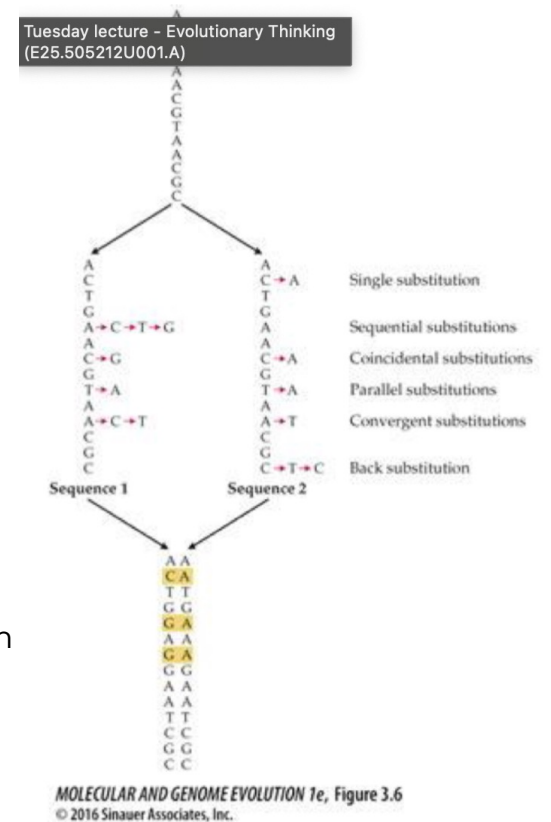
Divergent sequences

AGATGCTAGCATCGACTAGCARCAGCTGACCCGCGCGCGCAT  
AGAT**AA**TAGCATCGACTAGCARCAG**C**GGACCCGCGCGCGCAC**C**

You could just count how many difference there are but that's wrong

Mutations can hit the *same site twice* ( $A \rightarrow G \rightarrow A$ ). If that happens, the site looks unchanged, even occurred.

A site could also mutate back and forth several times.



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AGATGCTAGCATCGACTAGCARCAGCTGACCCGCGCGCGCAT

→ estimate the *true evolutionary distance* between sequences  
not just the observed one.



# SUBSTITUTION MODELS

**D-estimation** -> proportion of nucleotide sites that differ

**Jukes-Cantor** -> corrects for multiple & reverse changes

+ separate rates for **transition** (purines:  $A \leftrightarrow G$  or pyrimidines  $C \leftrightarrow T$ ) and **transversions**

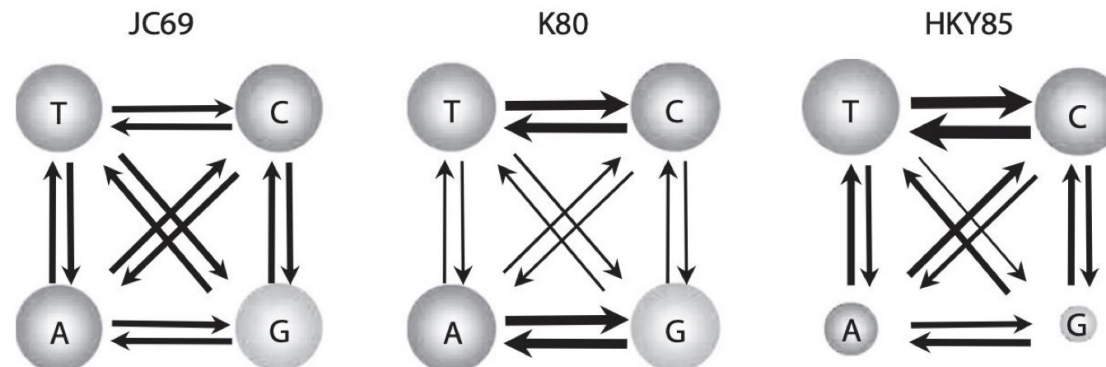
**Kimura's 2 parameter model**

+ Different equilibrium **frequency** distribution

**HKY85**

+ **variability** among sites

\*Model\* + **Gamma**

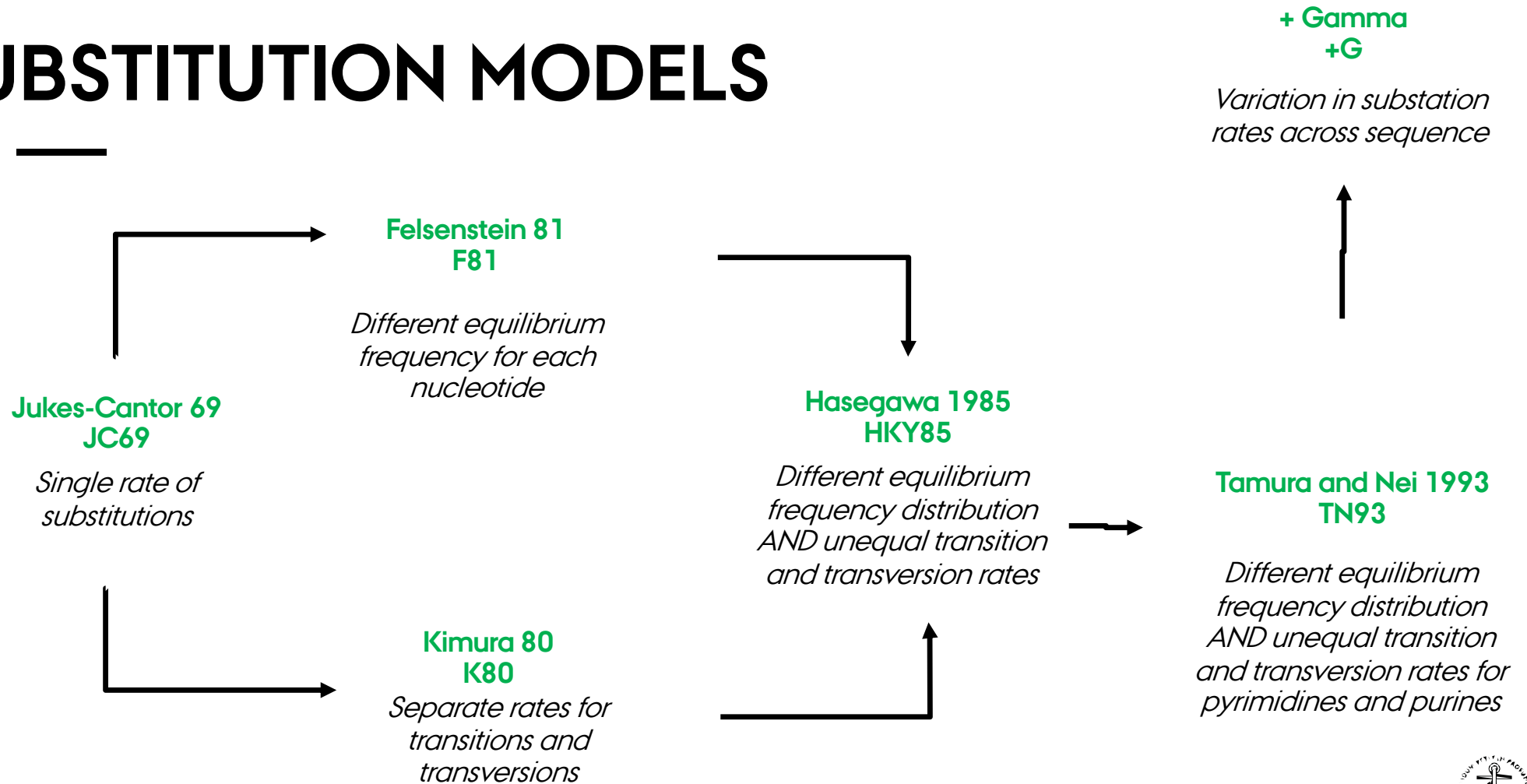


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# SUBSTITUTION MODELS





# WHICH MODEL IS BEST

These are “Nested model” -> one is a special case of the other model

## Simple models

- Few parameters (LOW sampling variance)
- Perform poorly for large distance

## Complex models

- Many parameters (LARGE sampling variance)
- Better behaved for large divergence
- Require LOTS of data

What is the true patterns of substitution ?



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# LIKELIHOOD FUNCTION

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Likelihood function

describes **the distribution of probabilities** that a model fitted to data describes it well

MLE (maximum likelihood estimation) – maximum value of the likelihood function for model describing studied data

- -> **highest possible probability** for the fitted model to really described the data

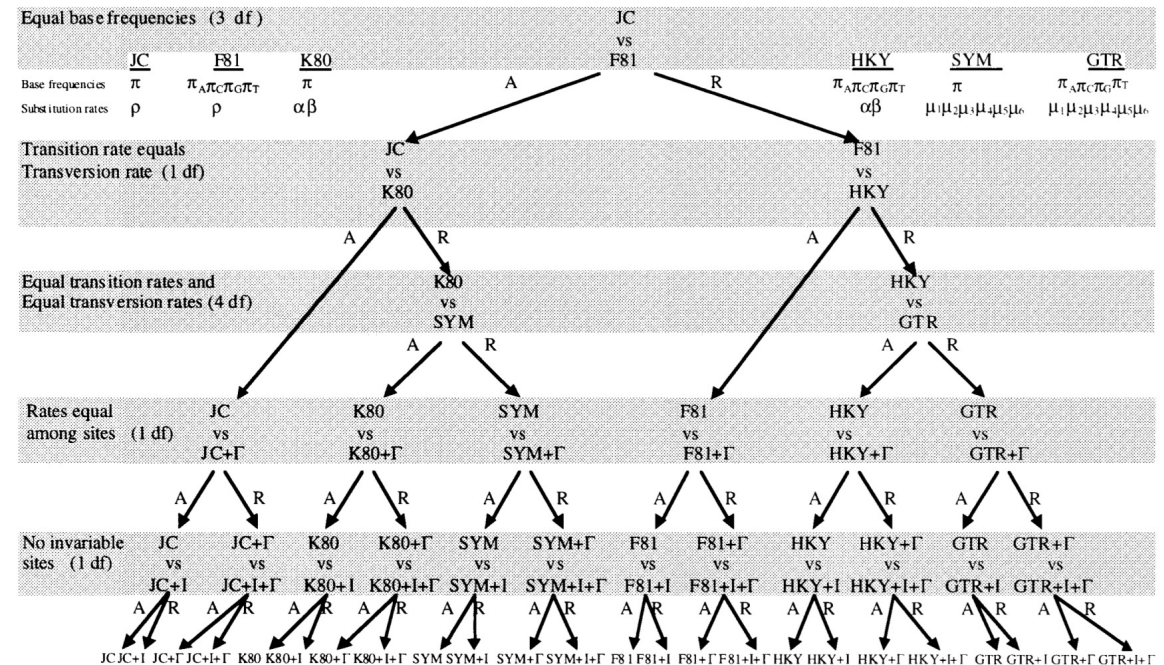
but more complex models are more prone to overfitting!

You test which model best fits your data (without overfitting) using the likelihood ratio test on nested pairs, checked against a  $\chi^2$  distribution.



# MODELTEST PAPER

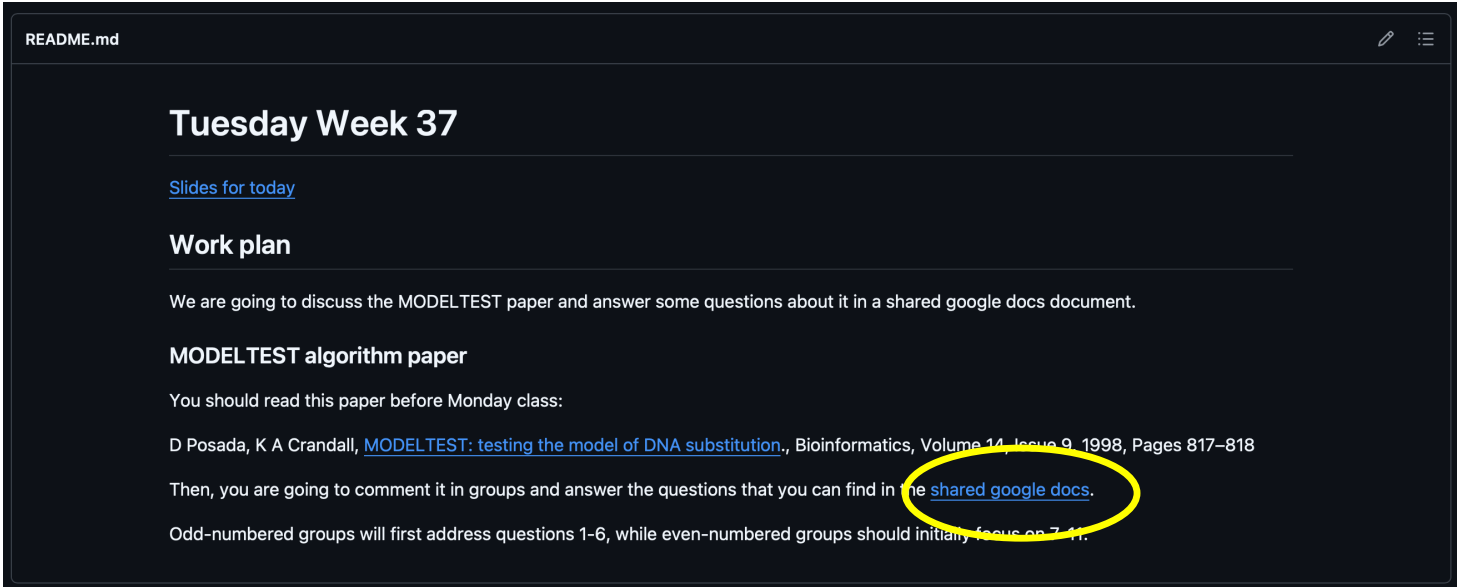
- Introduces an early solution for substitution model choice:  
simple model likelihood  
pairwise comparison
- Uses a likelihood ratio test to compare models
- Old but still a good early paper discussion resource



**Fig. 1.** Hierarchical hypothesis testing in MODELTEST. At each level the null hypothesis (upper model) is either accepted (A) or rejected (R). The models of DNA substitution are: JC (Jukes and Cantor, 1969), K80 (Kimura, 1980), SYM (Zharkikh, 1994), F81 (Felsenstein, 1981), HKY (Hasegawa *et al.*, 1985), and GTR (Rodríguez *et al.*, 1990).  $\Gamma$ : shape parameter of the gamma distribution; I: proportion of invariable sites. df: degrees of freedom. 1: equal base frequencies (0.25),  $\pi_A$ : frequency of adenine,  $\pi_C$ : frequency of cytosine,  $\pi_G$ : frequency of guanine,  $\pi_T$ : frequency of thymine.  $\rho$ : equal substitution rate,  $\alpha$ : transition rate,  $\beta$ : transversion rate;  $\mu_1$ : A $\Rightarrow$ C rate,  $\mu_2$ : A $\Rightarrow$ G rate,  $\mu_3$ : A $\Rightarrow$ T rate,  $\mu_4$ : C $\Rightarrow$ G rate,  $\mu_5$ : C $\Rightarrow$ T rate,  $\mu_6$ : G $\Rightarrow$ T rate.

# DISCUSSION OF THE PAPER – SEE THE GOOGLE DOCX

- Odd number groups will first answers questions 1-6
- Even-number groups will focus on answering question 7-11
- Half-way we will **mix** and you **explain your answers to each other**
- The questions are there to lead you through the paper



README.md

## Tuesday Week 37

[Slides for today](#)

### Work plan

We are going to discuss the MODELTEST paper and answer some questions about it in a shared google docs document.

### MODELTEST algorithm paper

You should read this paper before Monday class:

D Posada, K A Crandall, [MODELTEST: testing the model of DNA substitution.](#), Bioinformatics, Volume 14, Issue 9, 1998, Pages 817–818

Then, you are going to comment it in groups and answer the questions that you can find in the [shared google docs](#).

Odd-numbered groups will first address questions 1-6, while even-numbered groups should initially focus on 7-11.



# TOMORROW – MEGA EXERCISES

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MEGAX exercises: substitution models

Test different substitution models

Build phylogenetic trees :)



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